

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MEE 417 — Past Questions & Materials
by Renance

Course: MEE 417
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 400

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Paper 1 of 1 — MEE 417 OIL AND GAS INDUSTRY EQUIPMENT MAINTENANCE 2024/2025

Level 400 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

- COLLEGE OF TECHNOLOGY & ENGINEERING

DEPARTMENT OF MECHANICAL ENGINEERING

First Semester Examination 2024/2025 Session

Course & Title: MEE 417 Oil & Gas Industry Equipment Maintenance.

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Time Allowed: 2; Hours

Instruction: Answer five (5) questions,

(1a) Explain Static Equipment Engineering. what role does static equipment play in the oil and gas industry. (4 marks)

b, What is the difference between static and rotating equipment. (4 marks) -

c. A system is composed of five (5) components connected in series. Each component has an exponential time to failure

distribution with constant failure rate of 0.5 per 2,500 hours. Compute the reliability of the system for 1,500 hours of

operation and find MTTF. (6 marks)

2. For a system composed of five (5) elements in parallel, determine the system of reliability for 3000 hours of operation

and find MTTF. The five components have identical failure rate of 0.0005/hy and time to failure distribution exponential

in each case. What is MTTF of each component? (6 marks)

b. State four (4) best practices for oil rig maintenance management and what are the key components of an oil drilling rig.

(4 marks) PEsSeP

c. Explain workover operations and why is it necessary. (4 marks)

3a. With the aid of a diagram show the phases of Bath Tub curve failures and explain the phases. (6 marks)

3b. A particular machine has a constant failure rate of $\lambda = 0.05$ hours. What is the probability that it will fail within first 15

hours, (4 marks)- | X Siew al} &

3c. If the two reliability functions have same mean, show that their reliabilities may be different for the same operating

time, (4 marks) air Sit, of $(H < X_e \text{ SB pean Bay}$

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(4a) State five safety precautions for the maintenance of mechanical drives. (3 marks)

(4b) Calculate the transmission ratio between the driving wheel with 48 teeth and a pinion with 12 teeth of a machine

drive. (5 marks) 0. a F

(4c) Make a neat sketch of a distillation column, what kind of equipment would you classify it, (6 marks) wis) j

5 Sa. In a hand drill, the motor transmitted 3600 rpm to the drill spindle via two gears with 8 teeth and 32 teeth. Calculate

the rotational speed of the drill spindle. (5 marks)

Sb. What is the objective of accurate alignment? List four components that are most likely to fail. State five (5) results that

accurate alignment can achieve. (6 marks) ~~~? Sov Giburt refcebmitey aa defor, in ctthcyene

{ Sc. state five factors that are to be considered before any reliability test is being carried out. (3 marks)

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6. Explain four (4) types of lubrication application, state four (4) types of oils to avoid. (4 marks) > he f4

6b. Explain five types of maintenance strategies. | (5 marks) Pe re ee Seats

the five types of test you conduct on a lubricant. (5 marks) eg The P Gu. ¢ Wns \

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7a. State five (5) advantages of monitoring vibration. (5 marks) ~

b. What is the goal of maintenance and state two objectives of maintenance management. (3 marks)

¢. Calculate the effective reliability of a system in series having three sub-systems with reliability of 0.6, 0.9 and 0.8, if the

systems are now connected in parallel, what is the effective reliability. (6 marks)

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